Activity 02: Data Exploration

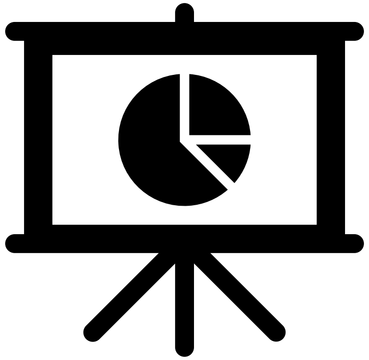
SISMID Lab

July 22 through July 24, 2026

A Human-AI Teaming Approach to Infectious Disease Modeling

Lecture Overview

General Housekeeping



Any slide with this icon contains the **overview** and **expected outputs** for the included tasks. All outputs listed on these slides should be uploaded to the course GitHub repository for review.

Any slide with this icon indicates that it is **time to complete a hands-on activity**. Carefully follow the instructions provided, complete the required task, and verify your results.

Lab Timeline

We are here

Activity 1
Data Cleaning & Visuals

Activity 2
Data Exploration

Activity 3
Modeling & Forecasting

Activity 4
Evaluation

Activity 5 Refinement

Task Overviews

Task	Description
Task 1: National Trend over Time (Q1)	Plot all U.S. influenza admissions from 2020 onward, highlight the 2025-2026 season with a gray shaded bar, and save a publication-ready *national_trend.png*.
Task 2: Season Comparison (Q2)	Overlay past seasons to compare trajectories, standardize dates across seasons, make the current 2025-2026 season stand out, and save a publication-ready *seasonal_comparison.png*.
Task 3: Peak Analysis, 2025-2026 (Q3)	Write the rules to find when the peak occurred and how large it was, when the decline started, and when the season began and ended. Output a *peak_description.csv* rather than a figure.

Goal Output

From this activity you should produce three outputs: a **national trend plot**, a **season comparison plot**, and a **peak-analysis table**. An AI agent generates them from the cleaned weekly admissions data, guided by your visualization rules.

Activity Output	Description
national_trend.png	National trend of U.S. influenza admissions from 2020 onward, with the 2025-2026 season highlighted.
seasonal_comparison.png	Past seasons overlaid on a standardized timeline, with the current 2025-2026 season emphasized.
peak_description.csv	A summary table of the 2025-2026 peak timing and size, when the decline started, and when the season began and ended.

Data Exploration & Visualization

Purpose: To explore the data before you model it and to write the rules that tell your agent what to plot and values to return.

What's different from Activity 1?

In Activity 1, the cleaning rules were provided for you.

In Activity 2, WE will write the rules.

`Remember: rules doc → agents.md → scripts/02_data_explore.R`

Task 1 (Q1): National Trend over Time Visualization

Q1: National Trend over Time Visualization

Plot all data available from the previous activity. As a reminder, this includes counts of total influenza admissions for the entire United States from 2020 onward obtained from the CDC (NHSN) data set.

Highlight the 2025-2026 season with a gray shaded bar.

Ensure that the figure is **publication-ready** (i.e., cleaned up) and is **saved** in the correct folder as ***national_trend.png.***

Writing the Visualization Rules

The rules should be specific enough that someone (or an AI) could produce the plot without asking any follow-up questions.

Things to think about:

Element	Description
Plot type	line chart, bar chart, scatter, etc.
X-Axis and Y-Axis	variable, label text, date format, start at zero?
Title	exact title text for the figure
Colors and highlights	line color, shading, annotation colors
Annotations and markers	peak markers, reference lines, text labels
Output file name	output/figures/02_data_explore/national_trend.png

Key Definition: What Is a "Season"?

We'll need this definition in the rules, so the agent knows how to identify season boundaries.

Influenza Season Boundaries

Time span: MMWR week 40 (≈ early October) through week 20 (≈ mid-May) of the next year. A season spans two calendar years and is named for both — e.g. 2025-2026.

Season start: First calendar date of first year with Epiweek = 40. This is a fixed calendar week, the same every season, so the agent sets it directly rather than detecting it from the data. Please account for any leap years in this determination.

Season end: Earliest calendar date in the second year with Epiweek = 20.

(CDC, Purpose & Methods, 2025)

Season-to-Date Mapping Note

When mapping calendar dates to MMWR epi-weeks, handle week numbers that fall in early January (for example epi-week 53) by using the calendar month to assign the season start year. Concretely, if a date's epiweek is >= 40 but its calendar month is January–August, attribute that date to the previous year's season (i.e., season_start_year = year - 1).

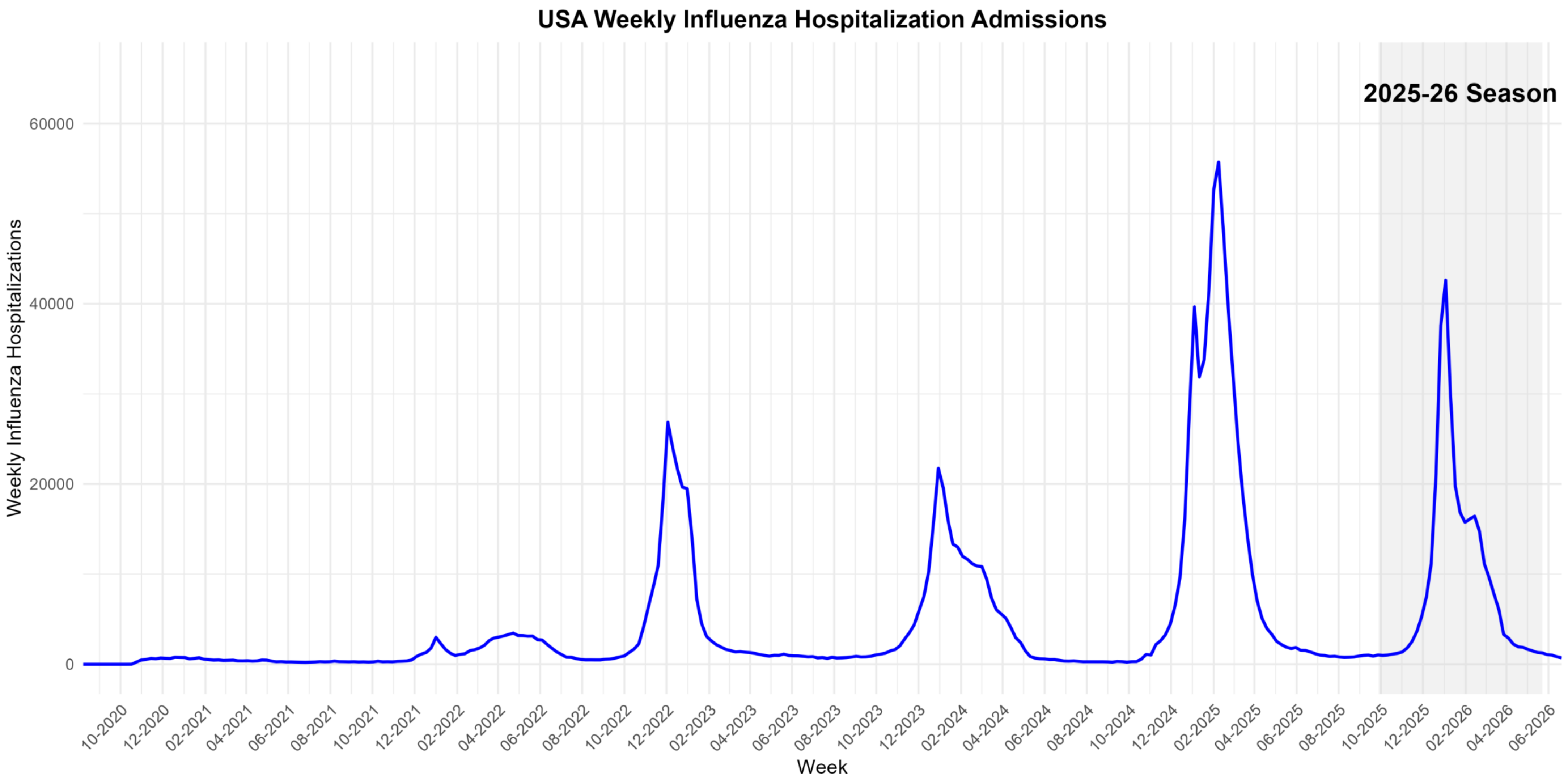
Hint: When specifying the season definition in the rules, copy-and-paste this directly in with the other specifications (previous slide).

Now it's your turn!

We will work through the following steps **together** to add the next section to our **rules.md** document and obtain **Q1**.

1. Create the `# Visualizations` section header in **rules.md**, and under that the `## National Plot` header.
2. Write detailed rules to obtain the visual for Q1. Make sure to specify input data details, season definitions, plot types, axes, colors, annotations, file names, etc. (vague rules = vague output).
3. Regenerate **agents.md** from your updated rules document
4. Use the **agents.md** file and Copilot (your AI of choice) to generate **scripts/02_data_explore.R.** Run the script and check the output.

national_trend.png



Task 2 (Q2): Season Comparison

Q2: Season Comparison

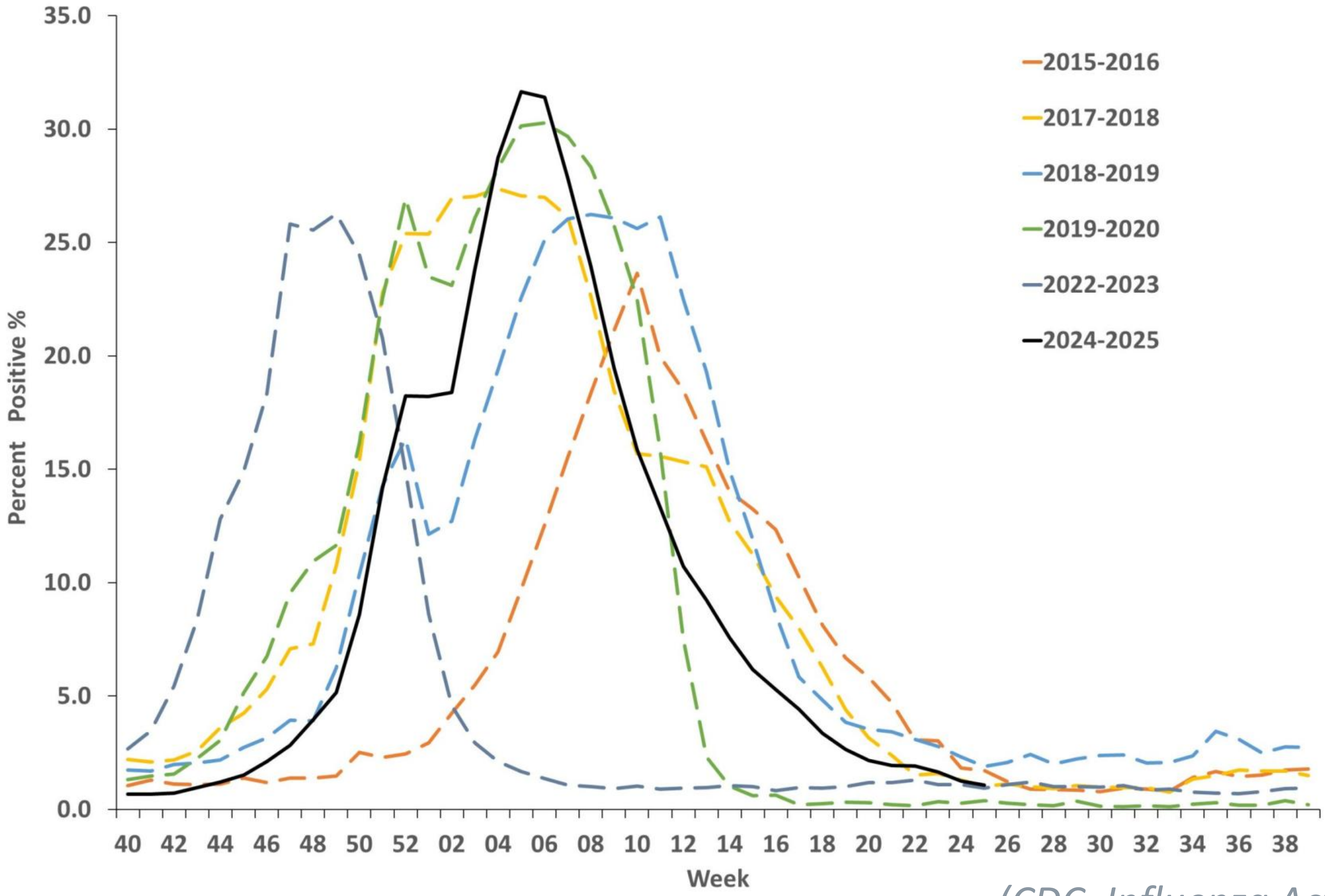
Plot all data available from the previous activity. Unlike Q1, we want to **overlay past seasons** to compare trajectories. Therefore, we will need to standardize dates across seasons (example on next slide).

Change the formatting of the lines so that the **current season (2025-2026) stands out**.

Ensure that the figure is **publication-ready** (i.e., cleaned up) and is **saved** in the correct folder as **seasonal_comparison.png.**

Example: Seasonal Comparison Plot

Style reference only.



(CDC, Influenza Activity in the US, 2026)

Writing the Visualization Rules

The rules should be specific enough that someone (or an AI) could produce the plot without asking any follow-up questions.

Things to think about:

Element	Description
Plot type	line chart, bar chart, scatter, etc.
Legend	Legend title, item labels, etc.
X-Axis and Y-Axis	variable, label text, date format, start at zero?
Title	exact title text for the figure
Colors and highlights	line color, shading, annotation colors
Annotations and markers	peak markers, reference lines, text labels
Output file name	output/figures/02_data_explore/seasonal_comparison.png

Key Definition: What Is a "Season"?

We'll need this definition in the rules, so the agent knows how to identify season boundaries.

Influenza Season Boundaries

Time span: MMWR week 40 (≈ early October) through week 20 (≈ mid-May) of the next year. A season spans two calendar years and is named for both — e.g. 2025-2026.

Season start: First calendar date of first year with Epiweek = 40. This is a fixed calendar week, the same every season, so the agent sets it directly rather than detecting it from the data. Please account for any leap years in this determination.

Season end: Earliest calendar date in the second year with Epiweek = 20.

(CDC, Purpose & Methods, 2025)

Plotting Note: The seasonal plot must present the full season chronologically by plotting MMWR weeks 40–53 followed by weeks 1–20 (so weeks 1–20 appear after 40–53 on the x-axis). The plotting implementation must ensure weeks 1–20 are not dropped and are positioned after weeks 40–53 so the full season displays continuously.

Hint: When specifying the season definition in the rules, copy-and-paste this directly in with the other specifications (previous slide).

Now it’s your turn!

We will work through the following steps **together** to add the next section to our **rules.md** document and obtain **Q2**.

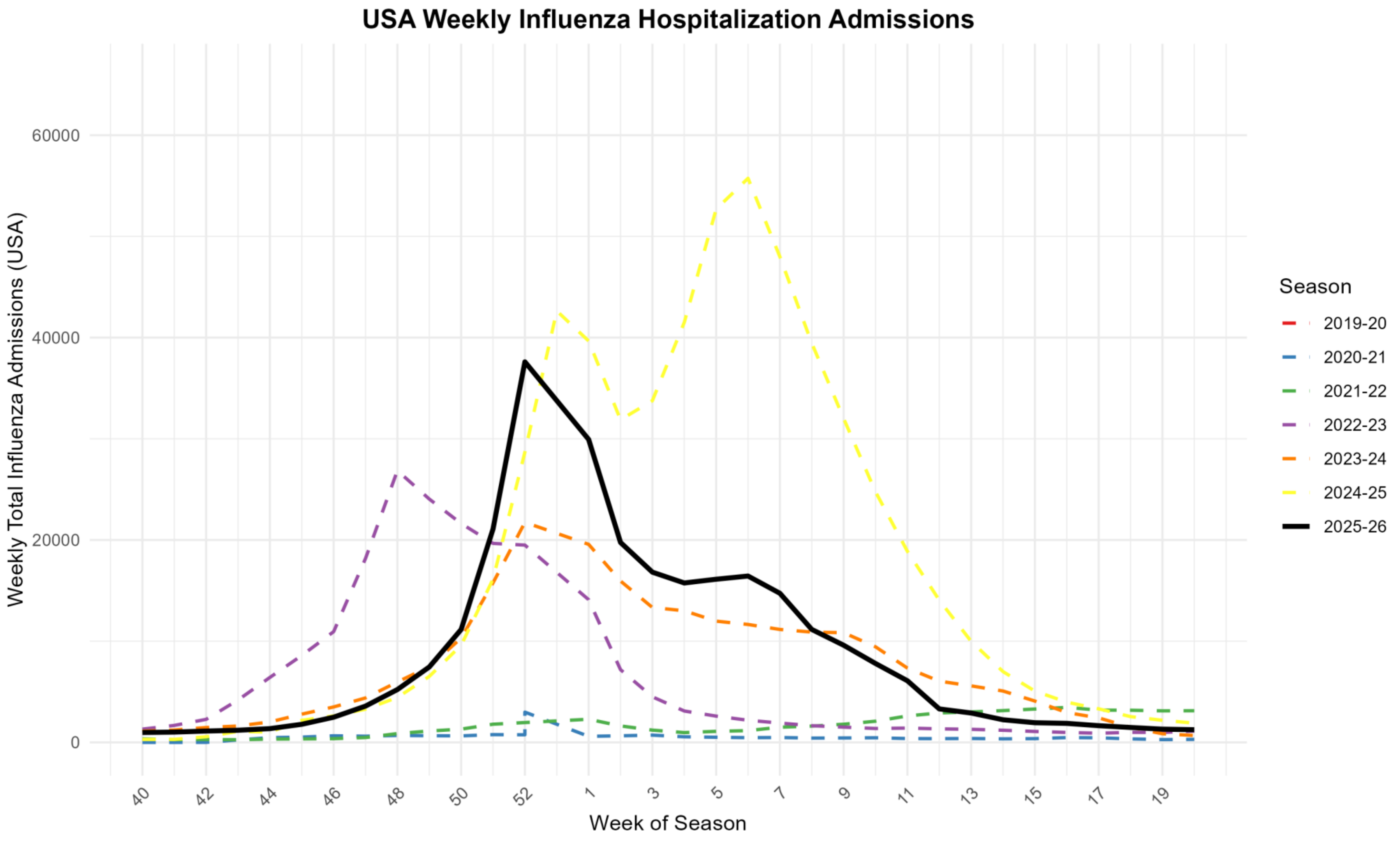
1. Create the `## Season Plot` section header under `# Visualizations` header in **rules.md**.
2. Write detailed rules to obtain the visual for Q2. Make sure to specify plot types, axes, colors, annotations, file names, etc. (vague rules = vague output).
3. Regenerate **agents.md** from your updated rules document
4. Use the **agents.md** file and Copilot (your AI of choice) to update **scripts/02_data_explore.R.** Run the script and check the output.

Now it’s your turn!

Now, **on your own**, update your rules to make the following changes to your figure and upload the final outputs to the course GitHub.

1. Increase the thickness of the current season plotted line and change the color of the line to blue.
2. Change the non-season plotted line styles. For example, if they are solid lines change it to dashed; if they are dashed, change it to solid.
3. Ensure that the x-axis label is “Week of Season” and the y-axis label is “Weekly Total Influenza Admissions (USA)”.
4. Upload the resulting rules document (Hint: Prompt Copilot, or AI of your choosing, to update the rules document with the new changes), **agents.md**, script and figure to GitHub. The figure should be named **seasonal_comparison.png**.

seasonal_comparison.png



Task 3 (Q3): Peak Analysis (2025-2026)

Q3: Peak Analysis (2025-2026)

Use the data created in Activity 1 to **answer the following questions** about the current season. Unlike Q1 and Q2, **you** will write the rules to obtain a **CSV** rather than a figure.

When was the peak and **how big** was it?
When did the **decline start**?

When did the **season start** and **end**?

Data File Format

Element	Description
Output file name	output/data/02_data_explore/peak_description.csv
Columns & Order	Peak_Time, Peak_Intensity, Decline_Start, Season_Start, Season_End
Variable Type	Date (YYYY-MM-DD): Peak_Time, Decline_Start, Season_Start, Season_End Numeric: Peak_Intensity

The GitHub agent will check for these columns and file name upon submission. Please make sure to **explicitly reference** this structure as you develop the rules to answer the peak questions.

Key Definition: What Is a "Season"?

We'll need this definition in the rules, so the agent knows how to identify season boundaries.

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Season start: First calendar date of first year with Epiweek = 40. This is a fixed calendar week, the same every season, so the agent sets it directly rather than detecting it from the data. Please account for any leap years in this determination.

Season end: Earliest calendar date in the second year with Epiweek = 20.

(CDC, Purpose & Methods, 2025)

Implementation Note

When computing `season_start_year` for each `week`, use the calendar year of the date (`year(week)`) as the base year passed to the assignment logic — do not use the `MMWRyear` returned by `MMWRweek()` as the base. This avoids mis-attribution when MMWR week numbers span calendar-year boundaries.

Hint: When specifying the season definition in the rules, copy-and-paste this directly in with the other specifications (previous slide).

Now it's your turn!

<u>**On your own**</u>, go through the following steps to add the next section to the **rules.md** document and obtain the answers for **Q3**. Once you are finished, **upload your results to GitHub**.

1. Create the `## Peak Analysis` section header under the `## Season Plot` header in **rules.md**.
2. Write detailed rules to obtain the CSV for Q3. Make sure to include where to call the data from, where to save, column headers and types, definition of a peak, etc.
3. Regenerate **agents.md** from your updated rules document
4. Use the **agents.md** file and Copilot (your AI of choice) to update **scripts/02_data_explore.R.** Run the script and check the output.

Remember, vague rules will result in vague output. Be precise about every detail!

peak_description.csv

Variable	Value
Peak_Time	2026-01-03
Peak_Intensity	42,634
Decline_Start	2026-01-10
Season_Start	2025-09-28
Season_End	2026-05-17